

第五章小结

系统分析

数学模型

指标

频率特性的定义:

$$G(j\omega) = \frac{|c_s(t)|}{|r(t)|} \{ \angle c_s(t) - \angle r(t) \}$$

$$= G(s)|_{s=j\omega} = \frac{F[c(t)]}{F[r(t)]}$$

典型环节的频率特性 $G_i(j\omega)$

系统的开环频率特性 $G(j\omega)$

Nichols 图线

$$\left\{ \begin{array}{l} \text{Nyquist 稳定判据} \\ \text{对数稳定判据} \end{array} \right\} \left\{ \begin{array}{l} Z = P - 2N \\ N = N^+ - N^- \end{array} \right.$$

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论

$$\text{低频段: } \left\{ \begin{array}{l} K_v \\ \Rightarrow e_{ss} = \frac{A}{K} \end{array} \right.$$

$$\text{中频段: 稳定裕度} \left\{ \begin{array}{l} \omega_c \rightarrow \gamma \\ \omega_g \rightarrow h \end{array} \right.$$

高频段:

$$\left\{ \begin{array}{l} \text{一阶 } t_s = 3T = \frac{3}{\omega_c} \\ \text{二阶 } \left\{ \begin{array}{l} \gamma \xrightarrow{P171 \text{ 图5-52}} \xi, \sigma\% \\ \omega_c \xrightarrow{P172 \text{ 图5-53}} t_s = \frac{7}{\omega_c \text{tg}\gamma} \end{array} \right. \\ \text{高阶 } \left\{ \begin{array}{l} \gamma \\ \omega_c \end{array} \xrightarrow{P173 \text{ 图5-56}} \left\{ \begin{array}{l} \sigma\% \\ t_s \end{array} \right. \end{array} \right.$$

准

快

抗高频
干扰能力

Nichols 图线

闭环系统的频率特性 $\Phi(j\omega)$

零频值 M_0

$$e_{ss} \xrightarrow{r(t)=1(t) \quad e(t)=r(t)-c(t)} 1 - M_0$$

谐振频率 ω_r

谐振峰值 M_r

带宽频率 ω_b

$$\left\{ \begin{array}{l} \text{一阶} \quad t_s = 3T = \frac{3}{\omega_b} \\ \text{二阶} \quad \left\{ \begin{array}{l} \omega_r = \omega_n \sqrt{1 - 2\xi^2} \\ M_r = \frac{1}{2\xi\sqrt{1 - \xi^2}} \end{array} \right\} \Rightarrow \left\{ \begin{array}{l} \xi \\ \omega_n \end{array} \right\} \Rightarrow \left\{ \begin{array}{l} \sigma\% \\ t_s \end{array} \right\} \\ \text{高阶} \quad \left\{ \begin{array}{l} M_r \\ \omega_r \end{array} \right\} \xrightarrow{P179 \text{ 图5-61}} \left\{ \begin{array}{l} \sigma\% \\ t_s \end{array} \right\} \end{array} \right.$$

准

快

系统校正

指标

数学模型

原系统性能 ~ 指标要求

